POLIMI GRADUATE SCHOOL OF MANAGEMENT

MILANO

Cloud Technologies and Big Data Frameworks

SUPERSTORE SALES DATASET ANALYSIS USING APACHE SPARK

Group 6

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AGENDA

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2. Dataset
3. Problem Definition
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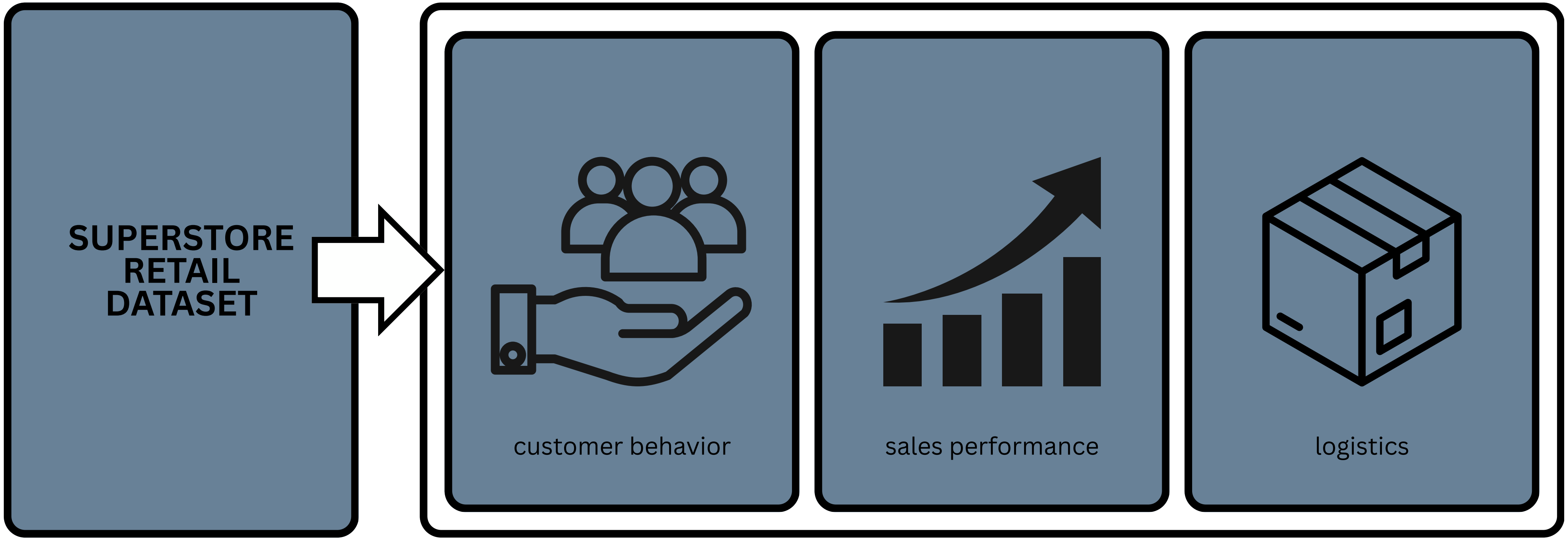
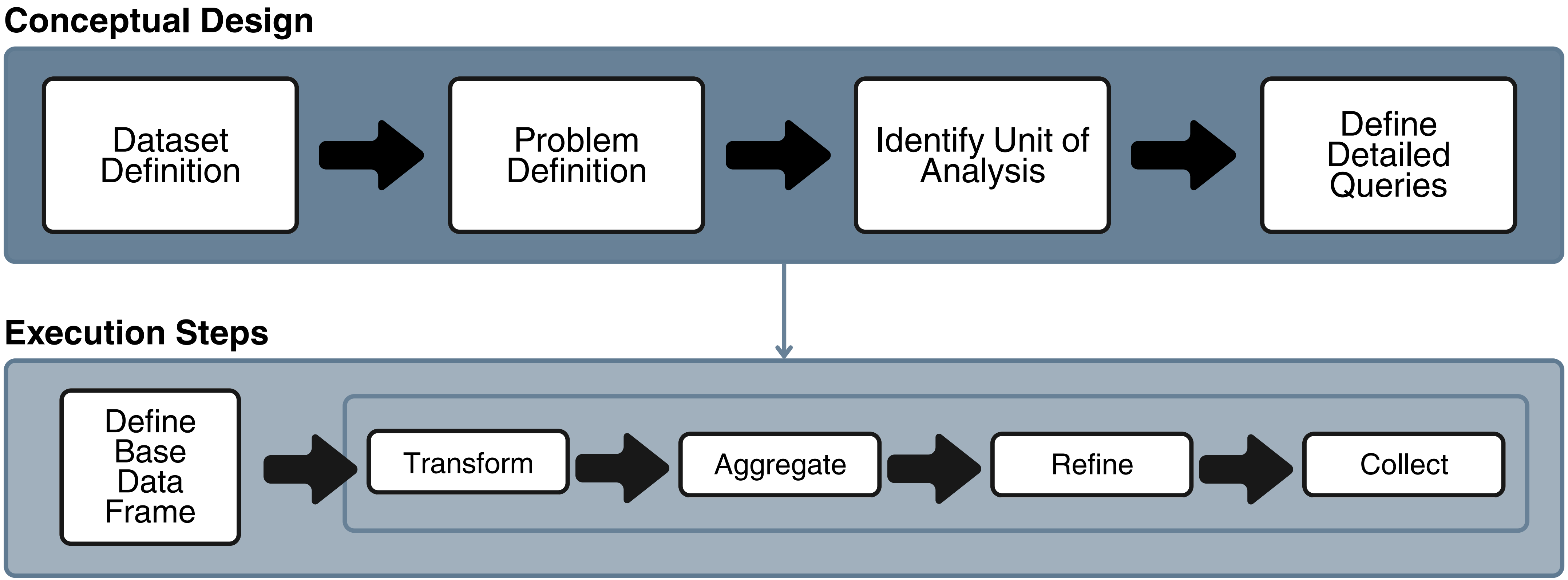
0. Project Overview



This project explores sales performance, customer behavior, and logistics performance of a SUPERSTORE using Apache Sparks DataFrame

1. Methodology

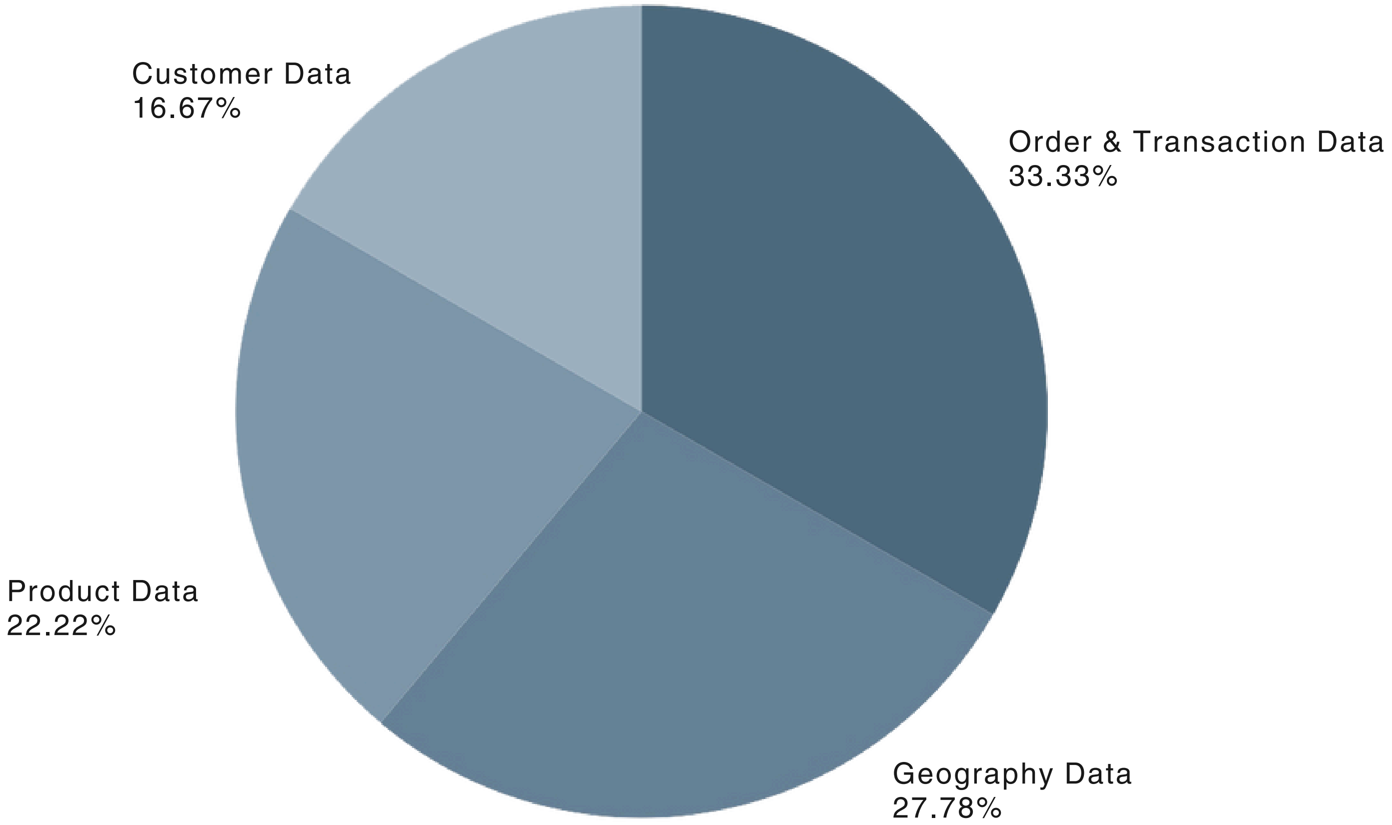


- First, the conceptual design defines the dataset, analytical problem, unit of analysis, and the specific questions to be answered.
- Second, the execution phase implements the analysis by constructing the base data frame for each queries and applying transformations, aggregations, and refinements to obtain the results.

2. Dataset

This project uses a transactional retail dataset from a global superstore, containing order-level information on sales, customers, products, shipping, and geographic locations.

Data Set Compositions:



Key Characteristics:

- Total Columns : 18 columns
- Total Rows : 9800 rows
- Time Range : 4 years

Detail Attributes:

1. Order & Transaction Data
    - Order ID
    - Order Date
    - Ship Date
    - Ship Mode
    - Sales
    - Row ID
2. Customer Data
    - Customer ID
    - Customer Name
    - Segment
3. Product Data
    - Product ID
    - Product Name
    - Category
    - Sub-Category
4. Geography Data
    - Country
    - Region
    - State
    - City
    - Postal Code

3. Problem Definition

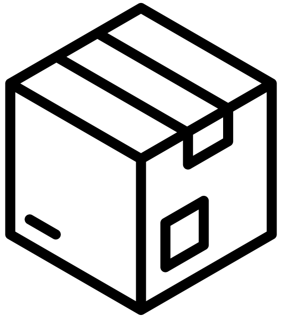
Unit of Analysis

Sales Performance

Revenue

Segment & Category

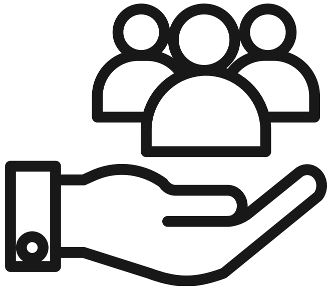
City & Time



Logistics

Shipping Duration

Shipping Delay



Customer Behaviour

Amount Spent

Purchase Frequency

Category Diversity

4. Data Frame

DF 1
MASTER VIEW

Raw transactional data → cleaned dates → unified base table

DF 2
MONTHLY GROWTH VIEW

DF1 → monthly sales by segment → month-over-month growth

DF 3
DAILY SALES VIEW

DF1 → assign day type (weekend/weekday)→ final join

DF 4
SHIPPING PERFORMANCE STATUS VIEW

DF1 → compute shipping delay using UDF → classify delivery status → aggregate city-level reliability metrics

DF 5
CUSTOMER VALUE & LOYALTY SEGMENTATION VIEW

DF1 → define customer feature→ compute value metrics → assign loyalty tiers based on spending amount and diversity

DF 6
CATEGORY REVENUE & PRICING PROFILE VIEW

DF1 → category aggregation → revenue & pricing metrics

4. Data Frame

4.1 Segmentation Rules Assumption

DF4 — Shipping Performance Status View

Assumptions Applied:

1. Delivery Speed Classification
    - Actual delivery duration is used to classify orders into Priority (0–2 days) and Standard (3+ days)
    - Enables comparison of fast vs standard delivery performance under a unified domestic benchmark
2. Late Delivery Definition by Ship Mode
    - Expected delivery time is defined using ship mode–specific maximum days
    - Orders exceeding these thresholds are labeled Late Delivery

DF5 — Customer Value & Loyalty Segmentation View

Assumption Applied:

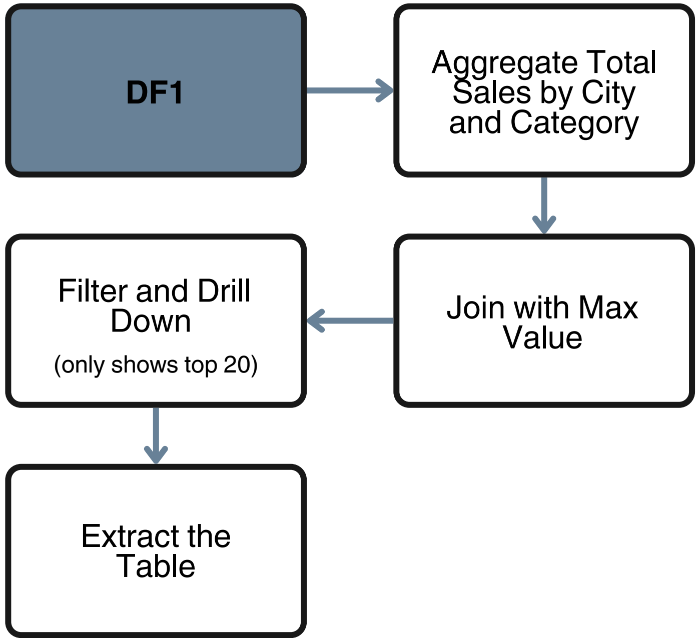
1. Customer Loyalty Tier Definition
    - Customers are classified using a fixed Lifetime Value threshold (≥ 3,000)
    - Category diversity is used to capture breadth of engagement across product categories

QUERIES

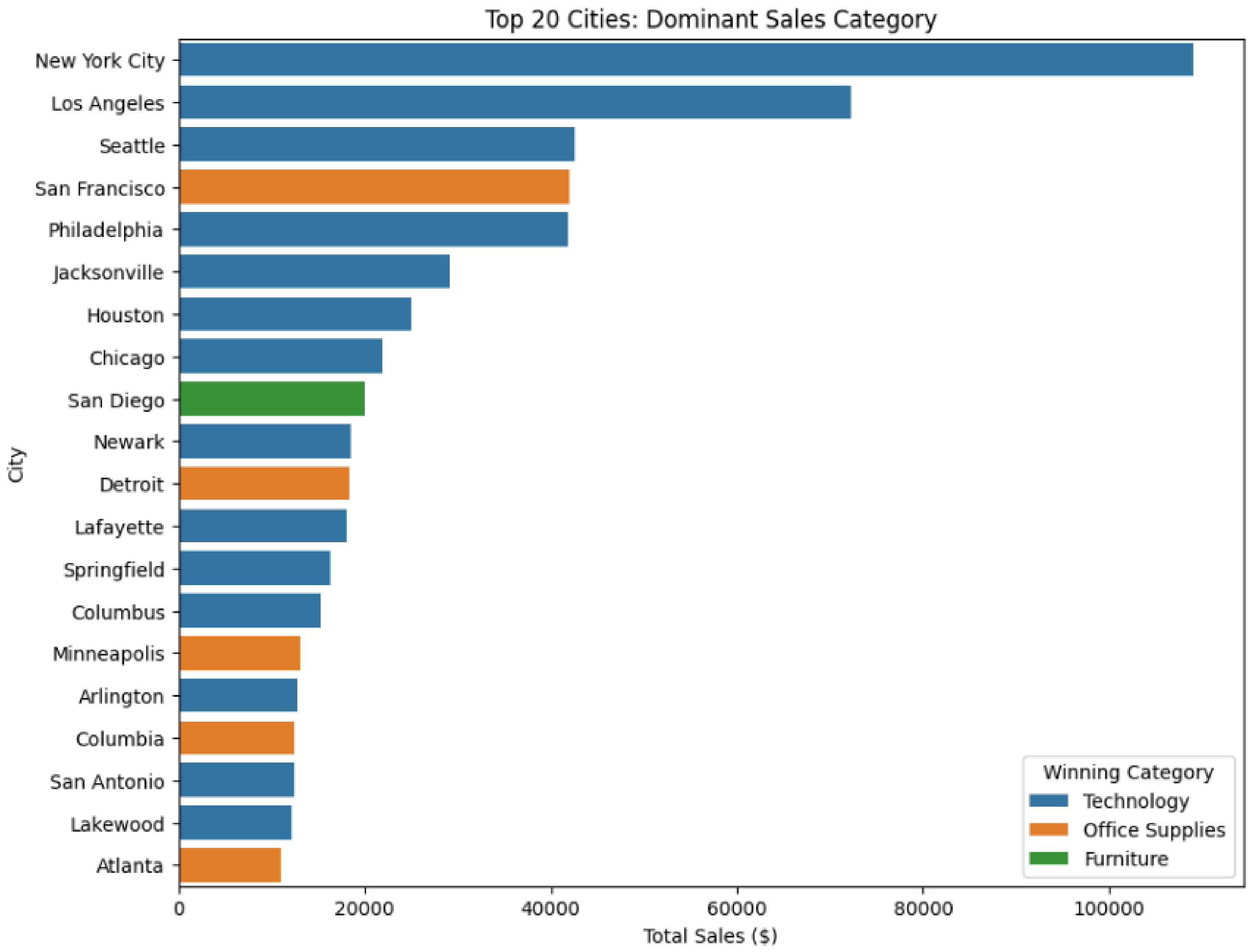
5.1 Query 1

Identify the dominant product category in each city

Steps:



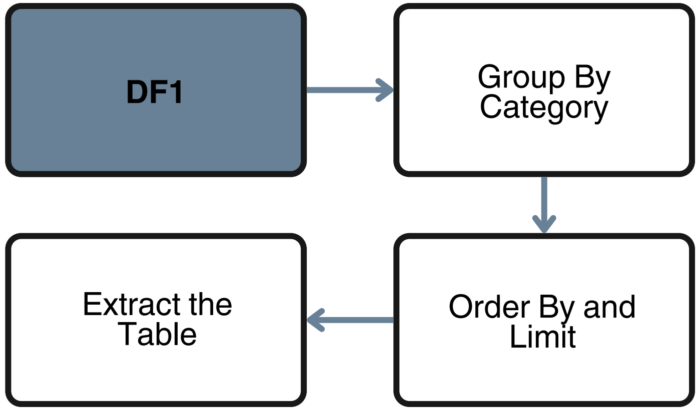
Result:



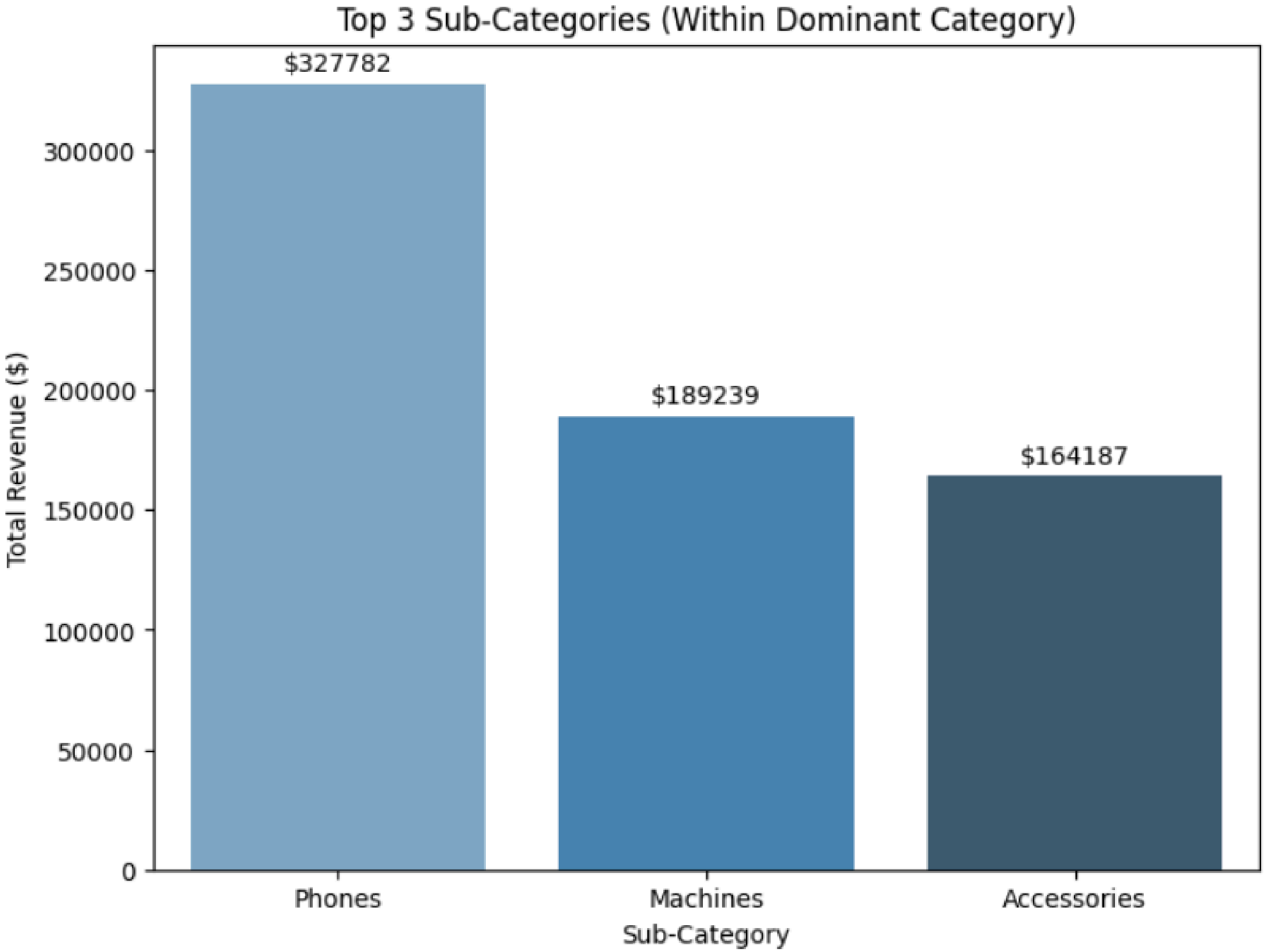
5.2 Query 2

Identify the top three sub-categories within the category with the highest total sales

Steps:



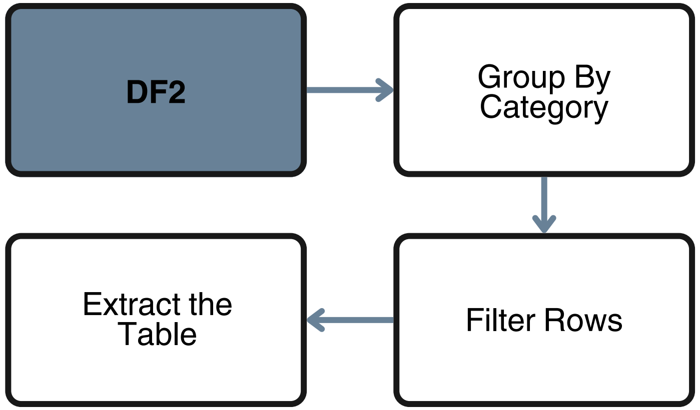
Result:



5.3 Query 3

Compute the average monthly revenue for Consumer customers

Steps:



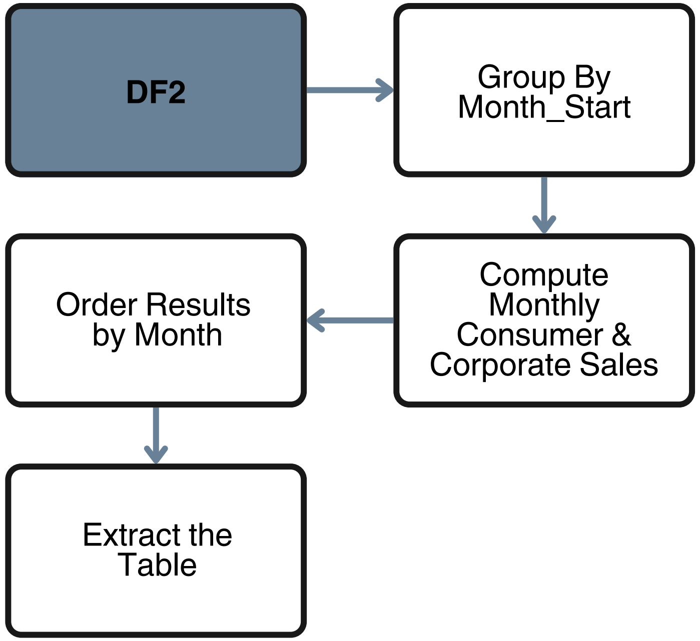
Result:

Average Monthly Revenue (monetary unit)
23,917.93

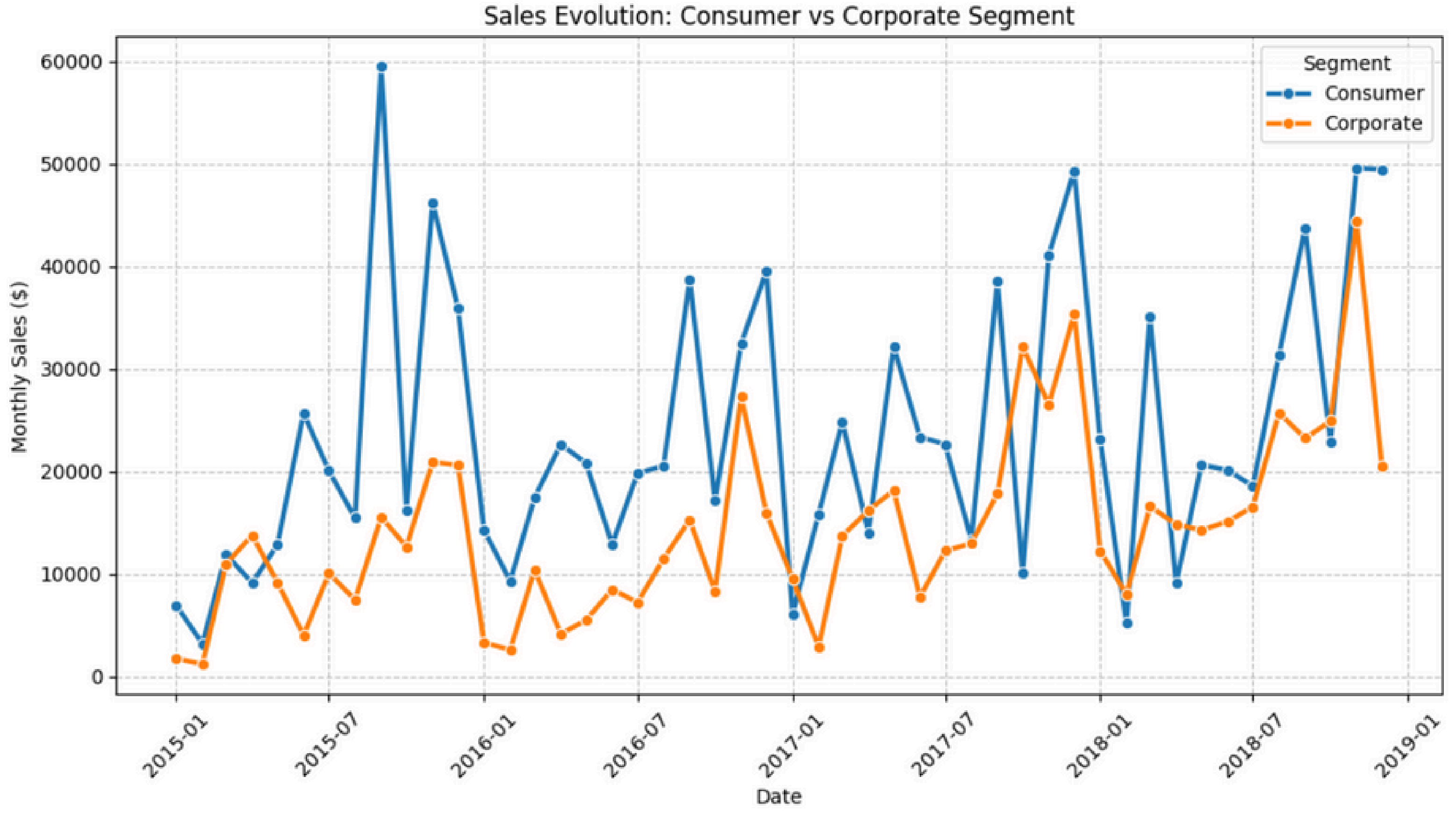
5.4 Query 4

Analyze the time evolution of sales, grouped by Consumer and Corporate segments

Steps:



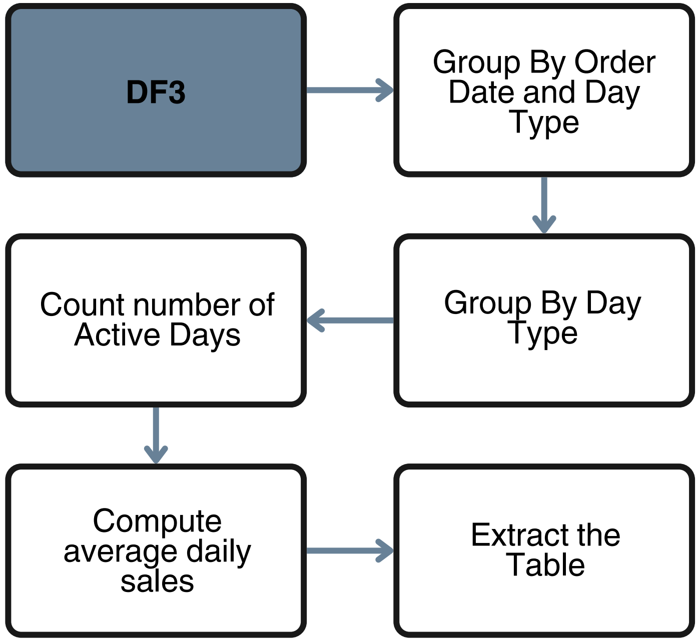
Result:



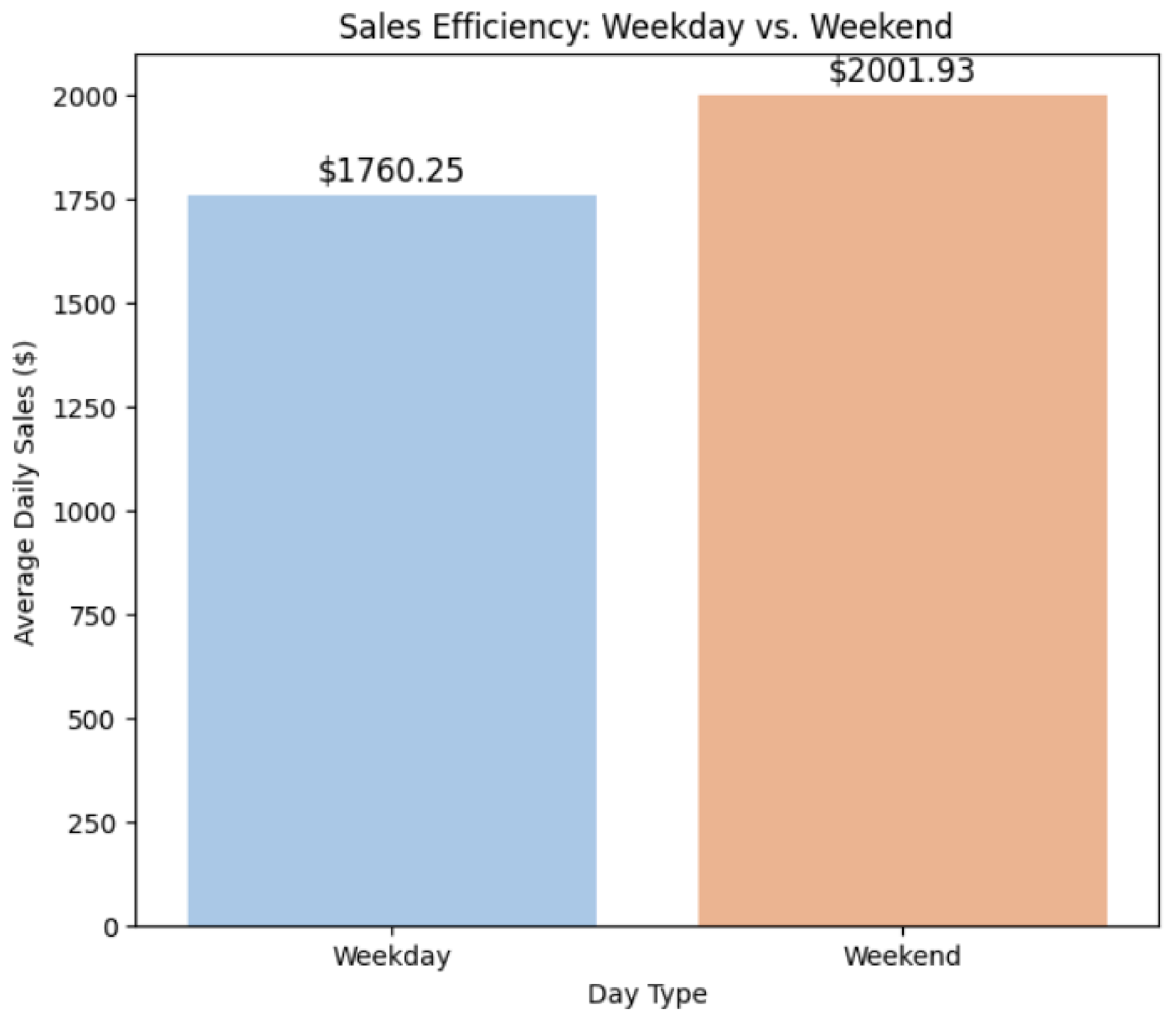
5.5 Query 5

Compare average daily sales efficiency between weekdays and weekends

Steps:



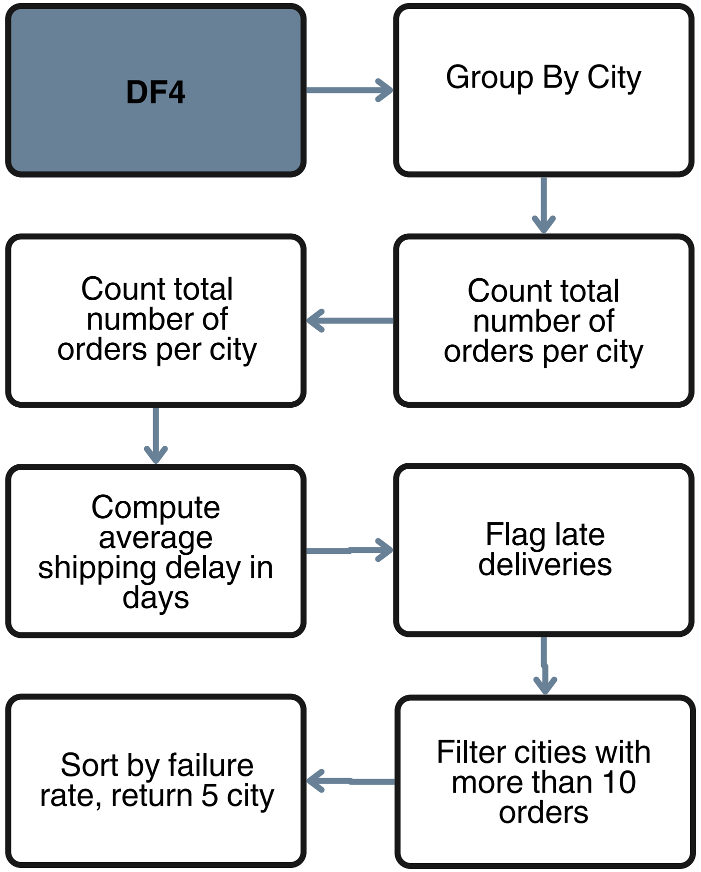
Result:



5.6 Query 6

Identify cities with the highest shipment failure rates (late deliveries)

Steps:

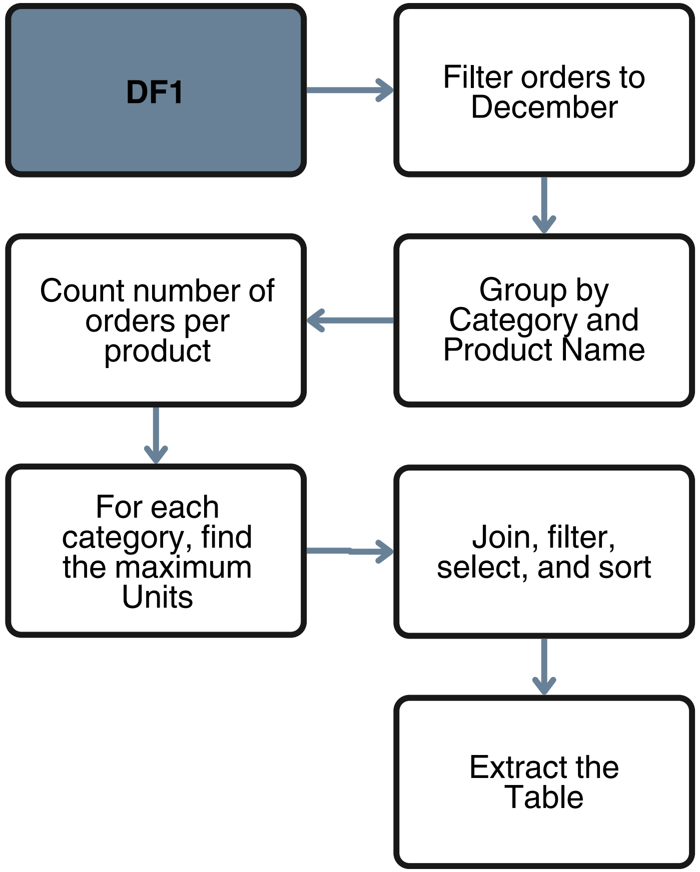


Result:

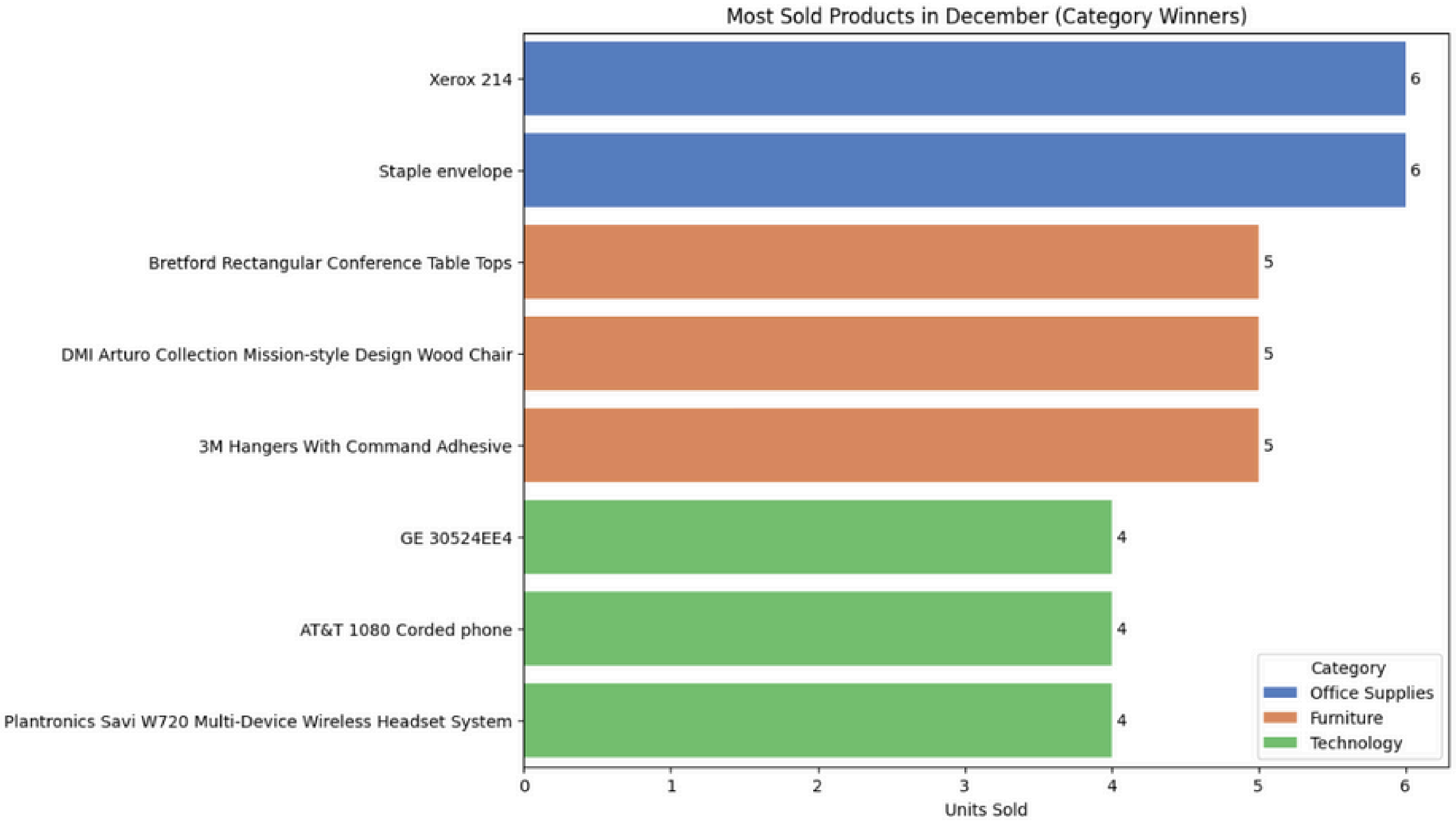
5.7 Query 7

For each category, determine the most sold product in the month of December

Steps:



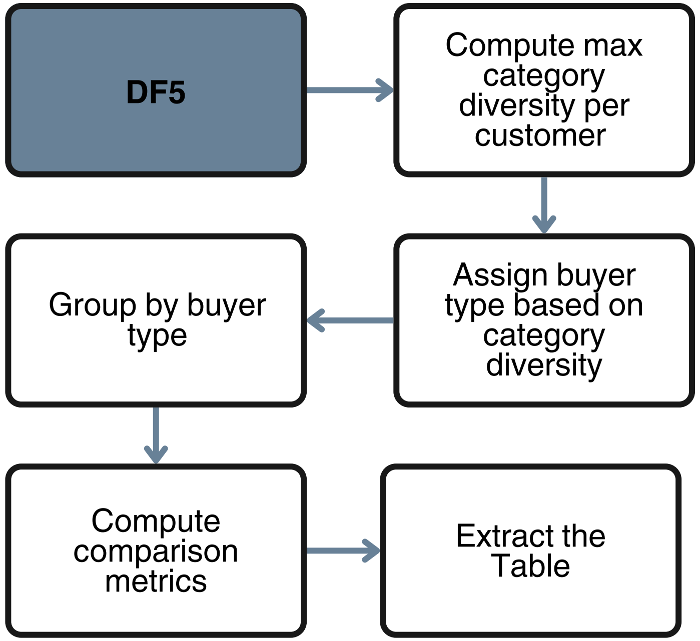
Result:



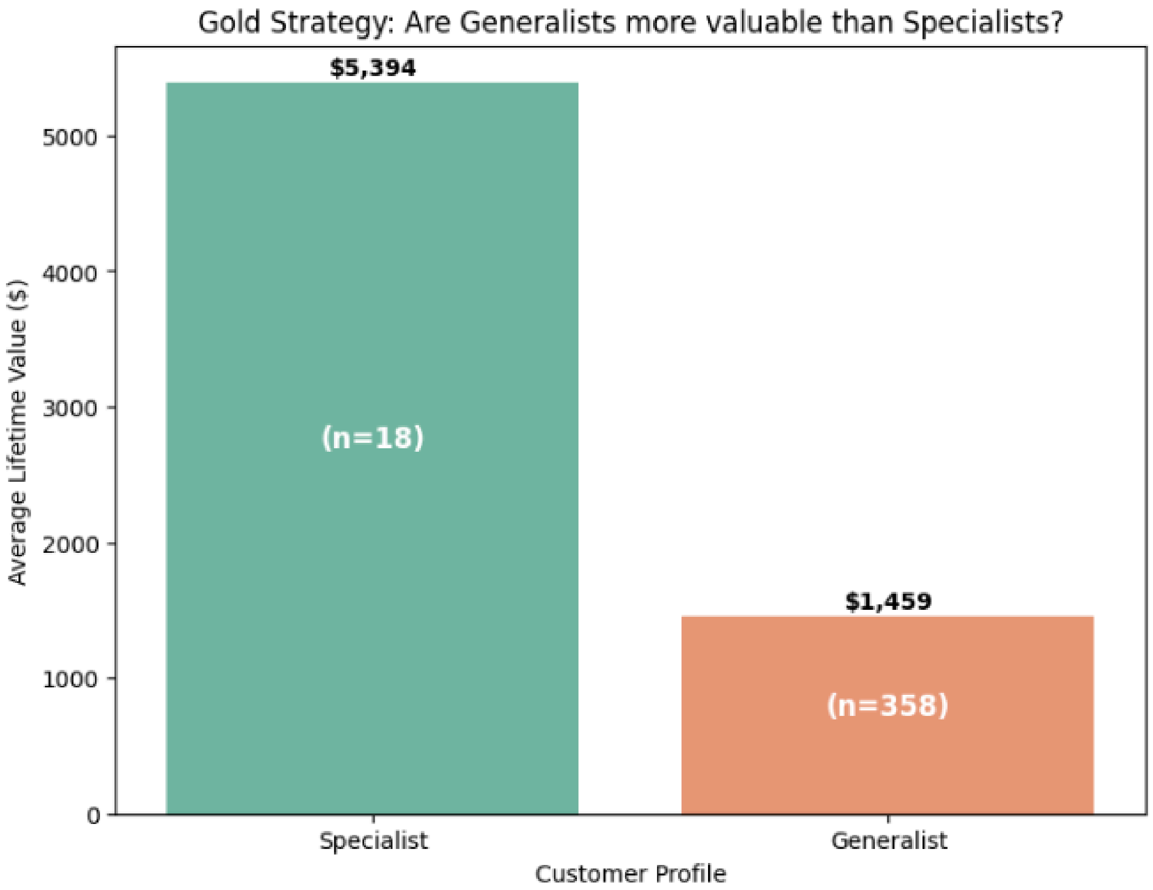
5.8 Query 8

Identify Gold customers who purchase from only one category (specialists) versus those who purchase across all categories (generalists)

Steps:



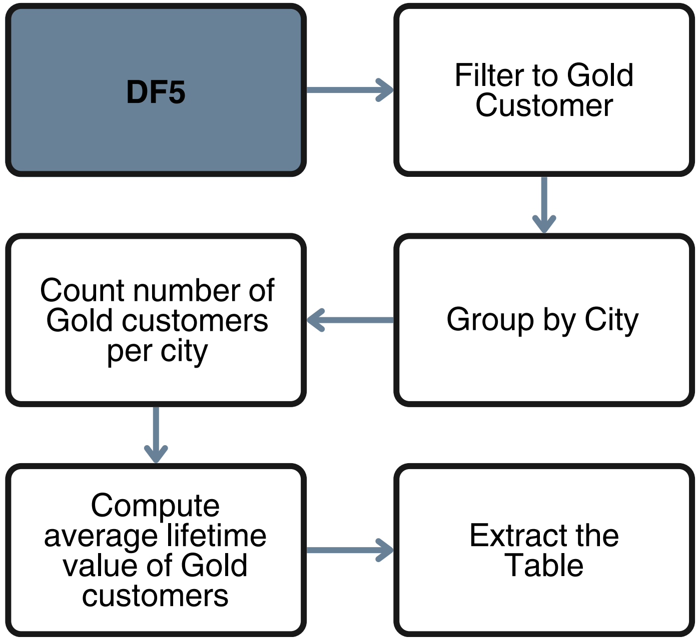
Result:



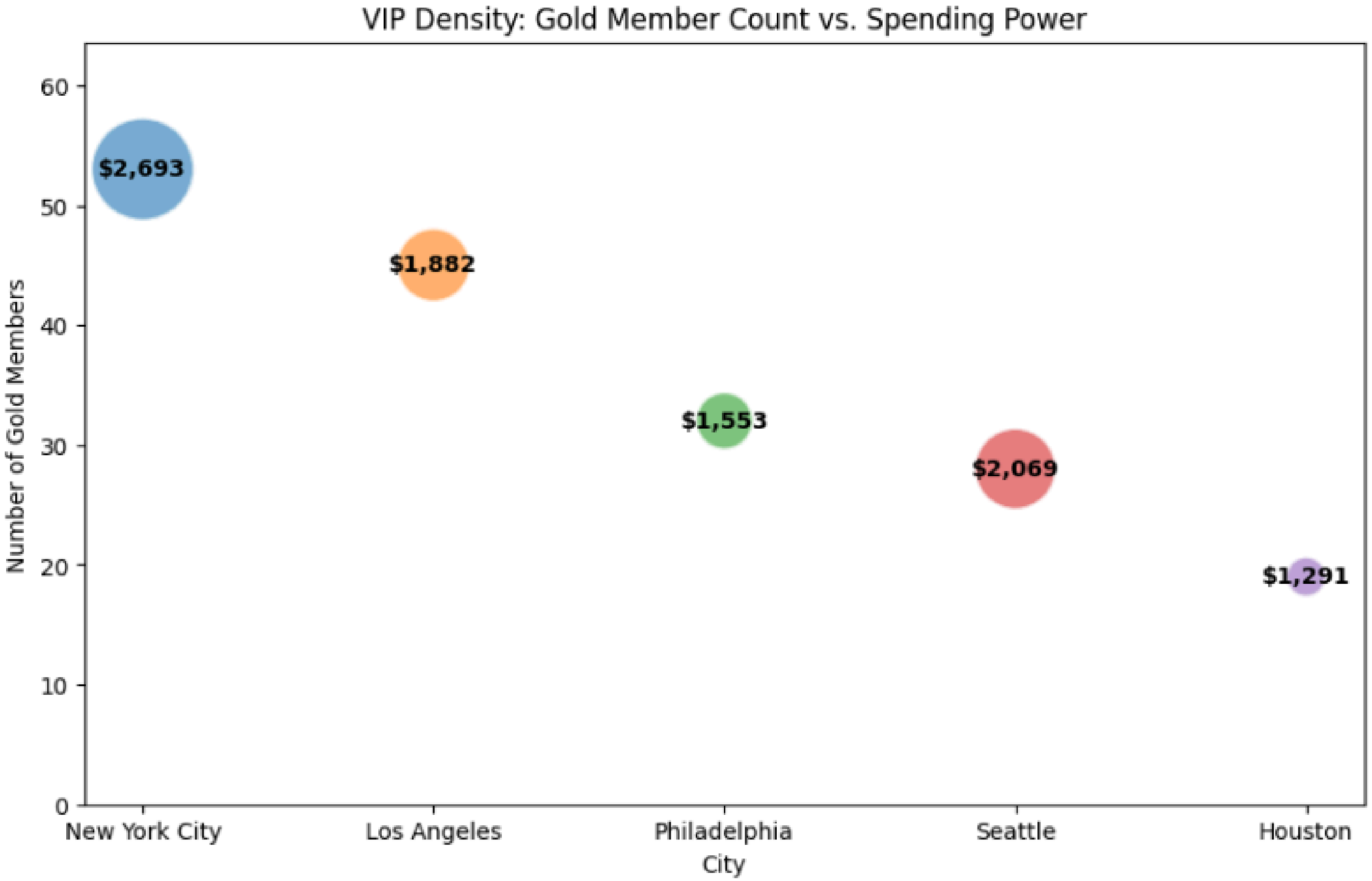
5.9 Query 9

Construct a City Wealth Index based on the concentration of Gold-tier customers

Steps:



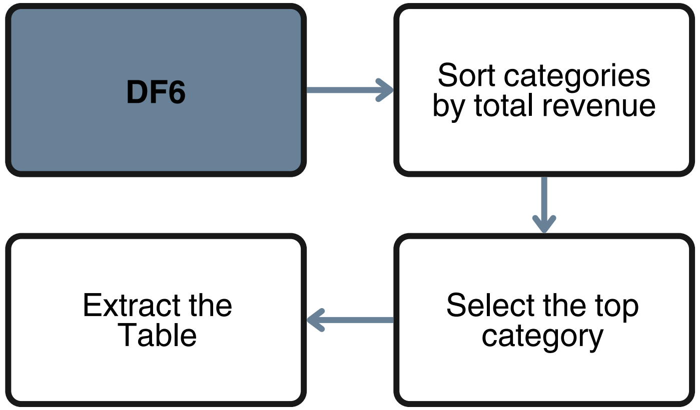
Result:



5.10 Query 10

Identify which product category generates the highest total revenue

Steps:



Result:

Category	Total Revenue	Average Ticket Size
Technology	827,455.87	456.4

THANK YOU